QUESTION ANALYSIS

EEE329

Communication Engineering I

Question Analysis

Sessions 2022–23 to 2019–20 *4 papers, 107 questions, 164 formulas*

INSIDE

• At a glance • Heat map • Priority list • 10 topic pages • Derivations • Repeats • Styles • Study order

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1. At a glance:

This analysis covers **4 papers** (2022-23, 2021-22, 2020-21, 2019-20), with 107 questions and 563 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Amplitude modulation	125	22 %	29	DSB-SC (43); AM modulation index (29); VSB (20)
Angle modulation	85	15 %	16	Deviation and index (28); Armstrong modulator (19); Carson's rule (9)
Multiplexing and multiple access	82	15 %	15	TDM (50); Multiple access (16); FDM (10)
Digital modulation	75	13 %	12	DPSK (22); ASK, FSK and PSK (16); QAM (14)
Sampling and pulse modulation	74	13 %	14	PCM (32); Delta modulation (26); Sampling (12)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2022-23	2021-22	2020-21	2019-20	Term tests	All
Signals and spectrum	·	**13**	**6**	**6**	**5**	**30**
Distortionless transmission	·	**6**	·	·	**5**	11
Propagation modes	·	**7**	·	·	·	7
Transmission media	·	·	·	**6**	·	6
EM spectrum	·	·	**6**	·	·	6
Noise and channel capacity	·	**6**	**20**	**13**	**10**	**49**
Channel capacity	·	**6**	**16**	**6**	·	28
Cascaded noise	·	·	·	**7**	**10**	17
Thermal and transit-time noise	·	·	**4**	·	·	4
Amplitude modulation	·	**40**	**17**	**23**	**45**	**125**
DSB-SC	·	**14**	·	**12**	**17**	43
AM modulation index	·	**15**	**4**	·	**10**	29
VSB	·	**6**	**7**	·	**7**	20
SSB	·	·	·	**11**	**6**	17
Envelope detector	·	**5**	·	·	**5**	10
Sidebands and power	·	·	**6**	·	·	6
Angle modulation	·	**10**	**19**	**36**	**20**	**85**
Deviation and index	·	**5**	**10**	**13**	·	28
Armstrong modulator	·	·	·	**9**	**10**	19
Carson's rule	·	·	**4**	·	**5**	9
FM bandwidth	·	·	·	**8**	·	8
FM demodulation	·	·	·	**6**	·	6
FM and PM	·	·	**5**	·	·	5
Pre-emphasis	·	·	·	·	**5**	5

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Topic and subtopic	2022-23	2021-22	2020-21	2019-20	Term tests	All
NBFM	·	**5**	·	·	·	5
FM versus AM	·	·	·	·	·	0
Receivers	·	·	**7**	·	·	**7**
Superheterodyne	·	·	**7**	·	·	7
Sampling and pulse modulation	·	**26**	**22**	**19**	**7**	**74**
PCM	·	**15**	**5**	**5**	**7**	32
Delta modulation	·	**6**	**10**	**10**	·	26
Sampling	·	**5**	**7**	·	·	12
Quantization	·	·	·	**4**	·	4
Digital modulation	·	**24**	**19**	**28**	**4**	**75**
DPSK	·	**6**	**4**	**8**	**4**	22
ASK, FSK and PSK	·	**11**	·	**5**	·	16
QAM	·	**7**	**7**	·	·	14
QPSK	·	·	·	**8**	·	8
FSK	·	·	**8**	·	·	8
Line coding	·	·	·	**7**	·	7
Multiplexing and multiple access	·	**18**	**28**	**17**	**19**	**82**
TDM	·	**12**	**7**	**12**	**19**	50
Multiple access	·	**6**	**10**	·	·	16
FDM	·	·	**5**	**5**	·	10
Spread spectrum	·	·	**6**	·	·	6
Information theory	·	**6**	**5**	**8**	**10**	**29**
Entropy	·	**6**	**5**	**8**	**10**	29
System design	·	**7**	·	·	·	**7**
Link design	·	**7**	·	·	·	7

3. Priority list:

Must know: Amplitude modulation (125), Angle modulation (85), Multiplexing and multiple access (82), Digital modulation (75)

Likely: Sampling and pulse modulation (74), Noise and channel capacity (49), Signals and spectrum (30), Information theory (29)

Less often: Receivers (7), System design (7)

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

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4. Topic pages:

4.1. Signals and spectrum:

30 marks, 5 questions, in 3 of 4 papers. Book: Lathi Ch. 1, 3; Forouzan Ch. 3.

Most often theory (3 of 5). Subtopics: Transmission media; EM spectrum; Distortionless transmission; Propagation modes.

Formulas:

Subtopic	Formula	Equation	Used in
Distortionless transmission (11)	**Distortionless channel**	$H(f) = k\, e^{-j2\pi f t_d}$	21-22 1(d), 20-21 7(b)
	Distortionless output	$y(t) = k\, x(t - t_d)$	21-22 1(d)
	Group delay	$t_g = -\frac{1}{2\pi}\frac{d\theta_h}{df} = t_d$	21-22 1(d)
Propagation modes (7)	Radio horizon (LOS)	$d \approx 4.12\left(\sqrt{h_t} + \sqrt{h_r}\right)$ km	21-22 1(b)

Symbols:

k = constant gain ↩ θ_h = phase response (rad) ↩ $\frac{4}{3}$-Earth radius ↩

t_d = constant delay (s) ↩ h_t, h_r = antenna heights (m),

Questions:

No.	Paper	Marks	Type	Question
Transmission media				
1	19-20 1(a)	6	theory	Write short notes on the following communication channels: (i) Twisted pair cable (ii) Radio wave
EM spectrum				
2	20-21 1(b)	6	theory	Briefly explain the three major bands of the electromagnetic (EM) spectrum-radio waves, microwaves and infra-red waves-in terms of their bandwidth, propagation mode and applications.
Distortionless transmission				
3	21-22 1(d)	6	derivation	Explain the conditions required for distortionless transmission.
4	20-21 TT1 7(b)	5	derivation	Explain distortion as a transmission impairment in communication systems. What are the conditions required for distortionless transmission?
Propagation modes				
5	21-22 1(b)	7	theory	Discuss the different modes of propagation of EM wave.

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4.2. Noise and channel capacity:

49 marks, 9 questions, in 3 of 4 papers. Book: Lathi Ch. 1, 13; Forouzan Ch. 3.

Most often numerical (5 of 9). Subtopics: Channel capacity; Cascaded noise; Thermal and transit-time noise.

Formulas:

Subtopic	Formula	Equation	Used in
Channel capacity (28)	**Shannon capacity**	$C = B\log_2\left(1+\frac{S}{N}\right)$ bit/s	21-22 1(c), 19-20 1(c)
	SNR for a rate	$\frac{S}{N} = 2^{C/B} - 1$	19-20 1(c)
	dB to ratio	$\text{SNR} = 10^{\text{SNR}_{dB}/10}$	19-20 1(c)
	Equally likely levels	$H = \log_2 L$	20-21 3(d)
	Information rate	$R = f_s H$ bit/s	20-21 3(d)
	Shannon capacity	$C = B\log_2\left(1+\frac{S}{N}\right)$	20-21 3(d)
	Decibels to ratio	$\frac{S}{N} = 10^{\text{dB}/10}$	20-21 5(b)
	Minimum SNR	$\text{SNR}_{min} = 2^{R/B} - 1$	20-21 5(b)
	Bandwidth for a rate	$B = \frac{R}{\log_2(1+S/N)}$	20-21 3(d)
	Attenuation	$20\log_{10}\frac{V_t}{V_r}$ dB	20-21 7(c)
	SNR from amplitudes	$\text{SNR} = (V_s/V_n)^2$	20-21 7(c)
	Signal for a rate	$V_s = V_n\sqrt{2^{R/B}-1}$	20-21 7(c)
	Attenuation	$10\log_{10}\frac{P_t}{P_r} = 20\log_{10}\frac{V_t}{V_r}$ dB	21-22 1(c)
	Signal-to-noise ratio	$\frac{S}{N} = \left(\frac{V_s}{V_n}\right)^2$	21-22 1(c)
	SNR for a rate	$\frac{S}{N} = 2^{R/B} - 1$	21-22 1(c)
Cascaded noise (17)	Thermal noise	$\overline{v_n^2} = 4kTBR$	19-20 1(d)
	Cascade noise resistance	$R_{eq} = R_1 + \frac{R_2}{A_1^2} + \frac{R_3}{A_1^2 A_2^2} + \cdots$	21-22 7(b), 20-21 7(d), 19-20 1(d)
Thermal and transit-time noise (4)	**Thermal noise power**	$P_n = kTB$	20-21 1(c)
	Thermal noise voltage	$V_n = \sqrt{4kTBR}$	20-21 1(c)

Symbols:

C = largest error-free rate (bit/s) ↩

B = channel bandwidth (Hz) ↩

L = number of levels ↩

f_s = symbols per second ↩

S/N = plain power ratio ↩

V_t, V_r = sent and received amplitude (V) ↩

R = wanted data rate (bit/s); resistance (Ω) ↩

P_t, P_r = sent and received power (W) ↩

V_s, V_n = signal and noise amplitude (V), same resistance ↩

T = temperature (K)

R_i = noise resistance at stage i's input

A_i = voltage gain of stage i

Questions:

No.	Paper	Marks	Type	Question
Channel capacity				
1	21-22 1(c)	6	numerical	We have a communication channel with a bandwidth of 5 MHz. A signal with an amplitude of 10 V is transmitted, and the receiver detects a signal with an amplitude of 2 V and noise with an amplitude of 5 mV. (i) Calculate the attenuation. (ii) Determine the maximum channel capacity. (iii) To achieve a data rate of 100 Mbit s^{-1}, find the required signal level.
2	20-21 3(d)	10	numerical	An analog signal having 4 kHz bandwidth is sampled at 1.25 times the Nyquist rate, and each sample is quantized into one of 256 equally likely level. Assuming the samples to be statistically independent: (i) What is the information rate of the source? (ii) Can the output of this source be transmitted without error over the AWGN channel with a bandwidth of 10 kHz and S/N ratio of 20 dB? (iii) Find S/N ratio required for error-free transmission of part ii. (iv) Find the bandwidth required for an AWGN channel for an error free transmission of output of this source if S/N ratio is 20 dB
3	20-21 5(b)	6	numerical	A voice grade telephone channel has a bandwidth of 3400 Hz. If the signal to noise ratio (SNR) on the channel is 30 dB, determine the capacity of the channel. If the above channel is to be used to transmit 4.8 kbps of data, determine the minimum SNR required on the channel.
4	20-21 TT1 7(c)	0	numerical	We have a communication channel with a bandwidth of 5 MHz. A signal with an amplitude of 10V is transmitted, and the receiver detects a signal with an amplitude of 2V and noise with an amplitude of 5 mV.
5	19-20 1(c)	6	numerical	States Shannon's capacity theorem. A user wishes to transmit data over a channel having bandwidth B = 3000 Hz. If the SNR is 9.08 find the theoretical maximum bit rate of the channel. Also find the required SNR if the desired bitrate of the channel is doubled.
Cascaded noise				
6	21-22 TT1 7(b)	5	derivation	Find the equivalent noise resistance for several amplifiers in cascade.
7	20-21 TT1 7(d)	5	derivation	Find the equivalent noise resistance for several amplifiers in cascade.
8	19-20 1(d)	7	derivation	Find the equivalent noise resistance for several amplifiers in cascade.
Thermal and transit-time noise				
9	20-21 1(c)	4	theory	Write short note on- (i) Thermal Noise (ii) Transit Time Noise

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4.3. Amplitude modulation:

125 marks, 29 questions, in 4 of 4 papers. Book: Lathi Ch. 4.

Most often theory (11 of 29). Subtopics: DSB-SC; SSB; AM modulation index; Sidebands and power; VSB; Envelope detector.

Formulas:

Subtopic	Formula	Equation	Used in
DSB-SC (43)	Product to sum	$\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$	19-20 1(b)
	DSB-SC of a tone	$a \cos \omega_m t \cos \omega_c t = \frac{a}{2}\cos(\omega_c - \omega_m)t + \frac{a}{2}\cos(\omega_c + \omega_m)t$	19-20 1(b)
	Square of a sine	$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$	19-20 1(b)
	rad/s to Hz	$f = \omega/2\pi$	19-20 1(b)
	Coherent detector output	$e_o(t) = \frac{1}{2}m(t)\cos(\Delta\omega t + \delta)$	19-20 2(c)
	DSB-SC spectrum	$S(f) = \frac{A_c}{2}[M(f-f_c) + M(f+f_c)]$	20-21 5(d)
	Coherent detector product	$V(f) = \frac{A_c}{2}M(f) + \frac{A_c}{4}[M(f-2f_c) + M(f+2f_c)]$	20-21 5(d)
	Lowest carrier	$f_c \geq W$	20-21 5(d)
	AM spectrum	$\pi A[\delta(\omega-\omega_c) + \delta(\omega+\omega_c)] + \frac{1}{2}[X(\omega-\omega_c) + X(\omega+\omega_c)]$	20-21 8(a)
	Coherent detector with errors	$e_o = [A + m(t)]\cos(\Delta\omega t + \delta)$	20-21 8(b)
	Coherent detection with errors	$e_o(t) = m(t)\cos(\Delta\omega\, t + \theta)$	21-22 2(b)
	Costas I and Q outputs	$\frac{1}{2}m(t)\cos\theta,\ \frac{1}{2}m(t)\sin\theta$	21-22 2(c)
	Costas control signal	$\frac{1}{8}m^2(t)\sin 2\theta$	21-22 2(c), 21-22 8(a)
	Square wave (± 1)	$w(t) = \frac{4}{\pi}\left[\cos\omega_c t - \frac{1}{3}\cos 3\omega_c t + \cdots\right]$	22-23 7(c)
	Ring modulator output	$\frac{4}{\pi}m(t)\cos\omega_c t$ after the BPF	22-23 7(c)
	DSB-SC with LO errors	$e_o = m(t)\cos(\Delta\omega t + \delta)$	22-23 7(e)
	SSB with LO errors	$e_o = m\cos(\Delta\omega t + \delta) - m_h \sin(\Delta\omega t + \delta)$	22-23 7(e)
SSB (17)	Sideband gap for filtering	$\text{gap} = 2f_L$	19-20 2(d)
	SSB (phase shift)	$\varphi_{SSB}(t) = m(t)\cos\omega_c t \mp m_h(t)\sin\omega_c t$	19-20 3(a)
	Hilbert transformer	$H(f) = -j\,\text{sgn}(f)$	19-20 3(a)
	SSB + carrier envelope	$\sqrt{(A+m)^2 + m_h^2} \approx A + m$	21-22 8(b)
	Bandwidth	AM, DSB-SC: $2B$; SSB: B	22-23 7(d)

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Subtopic	Formula	Equation	Used in
AM modulation index (29)	AM wave	$s(t) = [E_c + E_m \sin \omega_m t] \sin \omega_c t$	20-21 1(d)
	Modulation index	$\mu = \dfrac{E_m}{E_c}$	20-21 1(d)
	Side frequencies	$f_c \pm f_m$, amplitude $\dfrac{\mu E_c}{2}$	20-21 1(d)
	AM antenna current	$I_t = I_c\sqrt{1 + \mu^2/2}$	20-21 8(c)
	AM wave	$\varphi_{AM}(t) = [A + m(t)] \cos \omega_c t$	21-22 3(d)
	Modulation index	$\mu = \dfrac{m_p}{A}$	21-22 3(d)
	AM antenna current	$I_t = I_c\sqrt{1 + \dfrac{\mu^2}{2}}$	21-22 2(a)
Sidebands and power (6)	Carrier power	$P_c = \dfrac{E_c^2}{2R}$	22-23 7(b), 20-21 1(e)
	Total power	$P_t = P_c\left(1 + \dfrac{\mu^2}{2}\right)$	22-23 7(b), 21-22 2(a), 20-21 1(e)
	Modulation index	$\mu = E_m/E_c$	22-23 7(b)
	Side frequencies	$f_c \pm f_m$, amplitude $\mu E_c/2$	22-23 7(b)
	Efficiency	$\eta = \dfrac{\mu^2}{2 + \mu^2}$	22-23 7(b)
VSB (20)	**VSB receiver filter**	$H_o(f) = \dfrac{1}{H_i(f + f_c) + H_i(f - f_c)}$	21-22 2(d), 20-21 4(b)
	VSB signal	$\Phi_{VSB}(f) = [M(f + f_c) + M(f - f_c)]\, H_i(f)$	21-22 2(d)
	VSB filter condition	$H_i(f + f_c) + H_i(f - f_c) = \text{constant},\ \lvert f\rvert \le B$	21-22 2(d)
	VSB two tones	$\frac{1}{2}[a + (1 - a)] \cos \omega_1 t + \frac{1}{2} \cos \omega_2 t = \frac{1}{2} m(t)$	21-22 8(c)
	VSB filter condition	$H_i(f_c + f) + H_i(f_c - f) = 1$	22-23 7(g)
	VSB bandwidth	$B + f_v$	22-23 7(g)
Envelope detector (10)	Envelope detector RC	$\dfrac{1}{\omega_c} \ll RC < \dfrac{1}{2\pi B}$	21-22 3(b)
	Envelope detector RC	$\dfrac{1}{\omega_c} \ll RC \le \dfrac{\sqrt{1 - \mu^2}}{\mu \omega_m}$	21-22 7(d)

Symbols:

$\Delta\omega$ = local-carrier frequency error (rad/s)

δ = phase error

$M(f)$ = message spectrum ↩

W = message bandwidth ↩

$\Delta\omega$, δ = frequency and phase error ↩

θ = phase error (rad) ↩

f_L = lowest message frequency ↩

m_h = Hilbert transform ↩

E_c, E_m = peak carrier and message (V)

A = carrier amplitude ↩

ω_c = carrier frequency (rad/s) ↩

I_t = rms current with modulation (A) ↩

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I_c = rms current of the carrier alone (A) ↩

R = load (Ω) ↩

P_t = total power ↩

P_c = carrier power (W) ↩

E_m, E_c = message and carrier peaks ↩

H_i = transmit filter ↩

$H_o(f)$ = receiver LPF, for $|f| \leq B$ ↩

$H_i(f)$ = VSB shaping filter ↩

a = filter gain at $f_c + f_1$ ↩

f_v = vestige width ↩

RC = load time constant (s) ↩

Questions:

No.	Paper	Marks	Type	Question
DSB-SC				
1	22-23 TT2 7(c)	0	theory	Explain switching modulators, specifically the ring modulator: circuit, working, and the case $m(t) = 0$ where the carrier is suppressed.
2	22-23 TT2 7(e)	0	derivation	Explain the effect of local oscillator phase error and frequency error in coherent detection of DSB-SC/SSB.
3	21-22 2(b)	6	theory	Explain how phase and frequency error affect the coherent detection of AM.
4	21-22 2(c)	8	theory	With appropriate block diagram, briefly explain the working principle of Costas DSB-SC receiver.
5	21-22 TT2 8(a)	7	theory	What is quadrature null effect? Utilizing proper block diagram, explain principle of Costas receiver and how it prevents quadrature null effect.
6	20-21 5(d)	0	theory	Consider a message signal m(t) with the spectrum shown in Fig. 1. The message bandwidth W = 1 kHz. This signal is applied to a product modulator, together with a carrier wave $A_c \cos(2\pi f_c t)$, producing the DSBSC modulated wave s(t). This modulated wave is next applied to a coherent detector. Assuming perfect synchronism between the carrier waves in the modulator and detector, determine the spectrum of the detector output when: (i) the carrier frequency $f_c = 1.25$ kHz and (ii) the carrier frequency $f_c = 0.75$ kHz.
7	20-21 TT2 8(a)	5	diagram	Amplitude spectrum of a message signal shown in figure-1. Draw the spectrum of its DSB-WC and SSB-SC modulated signal.
8	20-21 TT2 8(b)	5	derivation	Explain how phase and frequency error affect the coherent detection of AM.
9	19-20 1(b)	6	numerical	For carrier frequency 100 kHz, find the frequency, sketch the spectrum, indicate the upper sideband, lower sideband and their amplitudes of the following signals. (i) $m(t) = 1 + \cos(377t)$ (ii) $m(t) = 2\sin^2(100\pi t)$ (iii) $m(t) = 3\cos(1000t)\cos(3000t)$
10	19-20 2(c)	6	derivation	Explain the effect of phase and frequency error during coherent detection of DSB-SC.
SSB				
11	22-23 TT2 7(d)	0	theory	Compare AM (DSB-WC), DSB-SC and SSB-SC in modulation, demodulation, power efficiency and bandwidth.
12	22-23 TT2 7(f)	0	theory	Disadvantages of Weaver's method of SSB generation.
13	21-22 TT2 8(b)	6	derivation	Explain how SSB can be detected using envelope detection method.
14	19-20 2(d)	3	theory	Describe the condition on message signal spectrum for SSB generation using selective filtering.
15	19-20 3(a)	8	derivation	Using proper block diagram explain phase shifting method of SSB generation.
AM modulation index				
16	21-22 2(a)	5	numerical	The antenna current of an AM transmitter is 8 A when only the carrier is sent. But it increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percentage of modulation changes to 0.8
17	21-22 3(d)	10	diagram	For the AM signal with $m(t)$ in Fig. 1 given below, sketch $\Phi_A M(t)$ for $\mu = 0.5, \mu = 1$, and $\mu = 1.5$.

No.	Paper	Marks	Type	Question
18	21-22 TT1 7(c)	5	numerical	The antenna current of an AM transmitter is 8 A when only the carrier is sent. But it increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percentage of modulation changes to 0.8.
19	20-21 1(d)	4	numerical	A carrier wave of frequency 10 MHz and peak value of 10 V is amplitude modulated by a 5 kHz sine wave of amplitude 6 V. Determine the modulation index, side frequencies and amplitude of side frequencies.
20	20-21 TT2 8(c)	5	numerical	The antenna current of an AM transmitter is 8 Amps, when only the carrier is sent, but it increases to 8.93A, when the carrier is modulated by a single sine wave. Find percentage modulation. Determine the antenna current when the percentage modulation changes to 0.8
Sidebands and power				
21	22-23 TT2 7(b)	0	numerical	An audio frequency signal $10\sin 2\pi 500t$ is used to amplitude modulate a carrier of $50\sin 2\pi 10^5 t$. Calculate: (i) Modulation index (ii) Sideband frequencies (iii) Amplitude of each sideband frequency (iv) Bandwidth required (v) Total power delivered to the load of 600 Ω (vi) Transmission efficiency
22	20-21 1(e)	6	numerical	An audio frequency signal $10\sin(2\pi 500t)$ is used to amplitude modulated a carrier of $50\sin(2\pi 105t)$ *[sic: should be* $2\pi 10^5 t$*]*. Calculate: (i) Modulation index (ii) Side-band frequencies (iii) Total power delivered to the load of 600 Ω
VSB				
23	22-23 TT2 7(g)	0	theory	What is VSB (vestigial sideband)? Why is it used?
24	22-23 TT2 7(h)	0	theory	Why is VSB-WC chosen for TV broadcasting?
25	21-22 2(d)	6	derivation	Find the condition of sideband shaping filter for VSB modulation.
26	21-22 TT2 8(c)	7	derivation	Using a message signal consists of two sinusoidal frequency components, describe the principle of VSB modulation and demodulation.
27	20-21 4(b)	7	derivation	Explain how the message signal is reconstructed from VSB signal.
Envelope detector				
28	21-22 3(b)	5	theory	Write a short note on the linear diode detection (envelope detector) method with proper diagrams and leveling.
29	21-22 TT1 7(d)	5	derivation	Explain the principle of envelope detector and hence find the RC requirement of envelope detector.

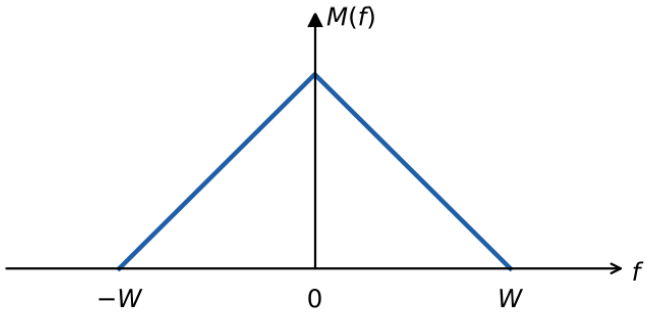


Figure 1: 20-21 5(d): Fig. 1 of the paper (redrawn)

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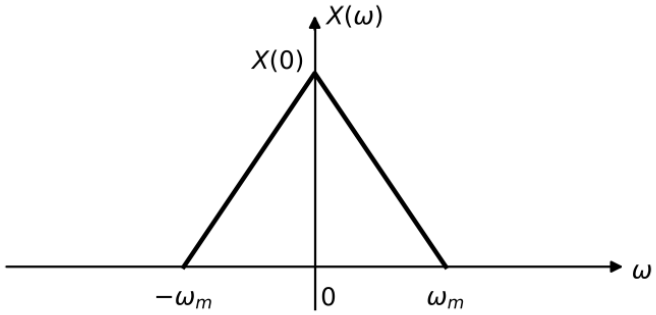


Figure 2: 20-21 TT2 8(a): Figure-1 of the paper (redrawn)

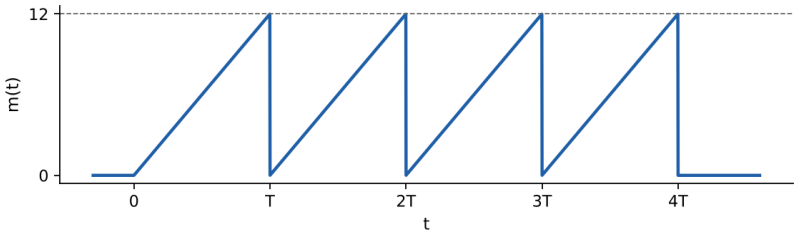


Figure 3: 21-22 3(d): Fig. 1 of the paper (redrawn): the message m(t)

4.4. Angle modulation:

85 marks, 16 questions, in 4 of 4 papers. Book: Lathi Ch. 5.

Most often theory (7 of 16). Subtopics: Deviation and index; FM bandwidth; Armstrong modulator; FM demodulation; FM and PM; Carson's rule; Pre-emphasis; NBFM; FM versus AM.

Formulas:

Subtopic	Formula	Equation	Used in
Deviation and index (28)	Instantaneous frequency	$\omega_i = \frac{d\theta}{dt},\ f_i = f_c + \frac{1}{2\pi}\frac{d\phi}{dt}$	19-20 2(b)
	Power of angle modulation	$P = \frac{A^2}{2}$	19-20 2(b)
	FM deviation	$\Delta f = k_f V_m$	20-21 2(d), 19-20 4(a)
	Modulation index	$\beta = \frac{\Delta f}{f_m}$ (or $\Delta f/B$)	19-20 4(a)
	Carson's rule	$B_T = 2(\Delta f + B) = 2B(\beta + 1)$	19-20 2(b)
	FM modulation index	$\beta = \frac{\Delta f}{f_m}$	21-22 3(c), 20-21 2(d)
	Frequency limits	$f_{max,min} = f_c \pm \Delta f$	21-22 3(c)
	Carrier swing	swing $= 2\Delta f$	21-22 3(c)
	Carson's rule	$B_T \approx 2(\Delta f + f_m)$	21-22 3(c)
FM bandwidth (8)	FM line spectrum	$A\sum_n J_n(\beta)\cos(\omega_c + n\omega_m)t$	19-20 3(b)
Armstrong modulator (19)	Multiplier	$f_c \to nf_c,\ \Delta f \to n\Delta f$	22-23 8(a), 19-20 3(c)

Subtopic	Formula	Equation	Used in
	Mixer	$f_c \to f_c \pm f_{LO}$, Δf unchanged	22-23 8(a), 19-20 3(c)
	Armstrong design	$n_1 n_2 = \dfrac{\Delta f}{\Delta f_1}$, $f_{LO} = \left\lvert \dfrac{f_c}{n_2} - n_1 f_{c1} \right\rvert$	22-23 8(a), 19-20 3(c)
FM demodulation (6)	Slope (differentiator) detector	$\dot{\varphi}_{FM} = -A[\omega_c + k_f m(t)]\sin(\cdot)$	19-20 4(b)
FM and PM (5)	Instantaneous frequency	$\omega_i = \dfrac{d\theta}{dt}$	20-21 2(c)
	PM	$\varphi_{PM} = A\cos[\omega_c t + k_p m(t)]$	20-21 2(c)
	FM	$\varphi_{FM} = A\cos\left[\omega_c t + k_f \int m\, d\alpha\right]$	20-21 2(c)
Carson's rule (9)	**Carson's rule**	$B_T \approx 2(\Delta f + f_m) = 2(\beta + 1) f_m$	20-21 4(d)
	Deviation from sensitivity	$\Delta f = k_f A_m$	20-21 8(e)
Pre-emphasis (5)	De-emphasis	$H_d(f) = \dfrac{1}{1 + jf/f_1}$, $f_1 = \dfrac{1}{2\pi RC}$	20-21 8(d)
NBFM (5)	FM wave	$\varphi_{FM}(t) = A\cos[\omega_c t + k_f a(t)]$, $a(t) = \int m(\tau)\, d\tau$	21-22 3(a)
	NBFM	$\varphi_{NBFM}(t) \approx A\cos\omega_c t - A\, k_f\, a(t)\sin\omega_c t$	21-22 3(a)
	FM signal	$A\cos[\omega_c t + k_f a(t)]$, $a(t) = \int m\, dt$	22-23 7(i)
	NBFM	$\varphi_{NBFM} \approx A\cos\omega_c t - A k_f a(t)\sin\omega_c t$	22-23 7(i)
	NBFM, tone	$A\cos\omega_c t + \frac{A\beta}{2}\cos(\omega_c + \omega_m)t - \frac{A\beta}{2}\cos(\omega_c - \omega_m)t$	22-23 7(i)
FM versus AM (0)	Carson's rule	$B_T = 2(\Delta f + B)$	22-23 7(a)

Symbols:

$\theta(t)$ = carrier angle

k_f = Hz per volt; frequency sensitivity (rad/s per volt); Hz/V ↩

B = highest message frequency ↩ ↩

f_m = modulating frequency (Hz) ↩

Δf = peak frequency deviation (Hz)

B_T = FM bandwidth (Hz) ↩

k_p = phase sensitivity ↩

Questions:

No.	Paper	Marks	Type	Question
Deviation and index				
1	21-22 3(c)	5	numerical	A 93.2 MHz carrier is frequency modulated by a 5 kHz sine wave. The resultant FM signal has a frequency deviation of 40 kHz. (i) Find the carrier swing of the FM signal. (ii) What are the highest and lowest frequencies attained by the frequency modulated signal? (iii) Calculate the modulation index for the wave.
2	20-21 2(d)	10	numerical	In an FM system, when the audio frequency is 500 Hz and modulating voltage 2.5 V, the deviation produced is 5 kHz. If the modulating voltage is now increased to 7.5 V, calculate the new value of frequency deviation produced. If the AF voltage is raised to 10 V while the modulating frequency dropped to 250 Hz, what is the frequency deviation? Calculate the modulation index in each case.

No.	Paper	Marks	Type	Question
3	19-20 2(b)	8	numerical	An angle modulated signal with carrier frequency 10 MHz is described by the equation $s(t) = 12\cos(\omega_c t + \sin 3000\pi t + 10\sin 2000\pi t)$ (i) Find the power of the modulated signal (ii) Find the frequency deviation Δf (iii) Find the β (iv) Find the bandwidth.
4	19-20 4(a)	5	numerical	In an FM system the frequency deviation is 5 kHz, when its audio modulating frequency is 400 Hz, and the audio modulating voltage is 2 V. Compute the modulation index. Also compute the frequency deviation and the modulation index if (i) if voltage is increased to 6V keeping modulation frequency unaltered, (ii) if voltage is increased to 8 V while the modulating frequency is reduced to 200 Hz.
FM bandwidth				
5	19-20 3(b)	8	theory	Explain the effect of variation of amplitude and frequency of message signal on the spectrum and bandwidth of FM.
Armstrong modulator				
6	22-23 TT3 8(a)	10	design	Design an Armstrong indirect FM modulator to generate an FM signal with carrier frequency 97.3MHz and $\Delta f = 20.48kHz$. A NBFM generator of $f_{c1} = 20kHz$ and $\Delta f = 5Hz$ is available. Only frequency doublers can be used as multipliers. Additionally, a local oscillator (LO) with adjustable frequency between 400 and 50-kHz *[sic: should be "500 kHz"]* is readily available for frequency mixing.
7	19-20 3(c)	9	design	Design an Armstrong indirect FM modulator to generate an FM signal with carrier frequency 97.3 MHz and $\Delta f = 10.24\ kHz$. A NBFM generator of $f_{c1} = 20\ kHz$ and $\Delta f = 5\ Hz$ is available. Only frequency doublers can be used as multipliers. Additionally, a local oscillator (LO) with adjustable frequency between 400 and 500 kHz is readily available for frequency mixing.
FM demodulation				
8	19-20 4(b)	6	theory	Explain slope detection method of FM demodulation.
FM and PM				
9	20-21 2(c)	5	theory	With proper block diagram explain the relation between frequency modulation and phase modulation.
Carson's rule				
10	20-21 4(d)	4	theory	State Carson's rule and its limitations.
11	20-21 TT2 8(e)	5	numerical	Consider an FM signal obtained from a modulating signal frequency of 2000 Hz and maximum Amplitude of 5 volts. The frequency sensitivity of modulator is 2 kHz/V. Find the bandwidth of the FM using Carson's rule.
Pre-emphasis				
12	20-21 TT2 8(d)	5	theory	Explain principle of preemphasis and deemphasis
NBFM				
13	22-23 TT2 7(i)	0	derivation	Derive the narrowband FM ($\beta < 1$) expression.
14	22-23 TT2 7(j)	0	diagram	Draw the NBFM block diagram.
15	21-22 3(a)	5	theory	Explain NBFM generation process with necessary block diagrams and equations.
FM versus AM				
16	22-23 TT2 7(a)	0	theory	Advantages and disadvantages of FM over AM.

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4.5. Receivers:

7 marks, 1 question, in 1 of 4 papers. Book: Lathi Ch. 4.

Most often theory (1 of 1). Subtopics: Superheterodyne.

Formulas:

Subtopic	Formula	Equation	Used in
Superheterodyne (7)	**Local oscillator**	$f_{LO} = f_s + f_{IF}$	20-21 4(a)
	Image frequency	$f_{si} = f_s + 2f_{IF}$	20-21 4(a)

Questions:

No.	Paper	Marks	Type	Question
Superheterodyne				
1	20-21 4(a)	7	theory	With appropriate block diagram, describe the operation of superheterodyne receiver.

4.6. Sampling and pulse modulation:

74 marks, 14 questions, in 3 of 4 papers. Book: Lathi Ch. 6.

Most often theory (7 of 14). Subtopics: Quantization; PCM; Delta modulation; Sampling.

Formulas:

Subtopic	Formula	Equation	Used in
Quantization (4)	Step size	$\Delta = \dfrac{2m_p}{L}$	21-22 4(e), 19-20 4(c)
	Quantization noise	$\dfrac{\Delta^2}{12} = \dfrac{m_p^2}{3L^2}$	19-20 4(c)
	SQNR (uniform)	$\dfrac{S_o}{N_o} = 3L^2 \dfrac{\overline{m^2}}{m_p^2}$	19-20 4(c)
	μ-law compressor	$y = \dfrac{\ln(1+\mu\lvert m\rvert/m_p)}{\ln(1+\mu)}\operatorname{sgn}(m)$	21-22 4(c), 19-20 4(c)
PCM (32)	Sampling rate	$f_s = (1+x)\,2B$	19-20 4(d)
	PCM bit rate	$R = (1+e)\,N_{sig} f_s n$	19-20 4(d)
	Minimum bandwidth	$B_T = R/2$	19-20 4(d)
	Bits per sample	$n = \log_2 L$	20-21 3(c)
	PCM bit rate	$R_b = n f_s$	21-22 4(e), 20-21 3(c)
	Information rate	$R = r\,H \quad \text{bits/s}$	21-22 4(c)
	Bits per sample	$L = 2^n,\ n = \log_2 L$	21-22 4(e)
	Quantization noise	$N_q = \dfrac{\Delta^2}{12} = \dfrac{m_p^2}{3L^2}$	21-22 4(a)
	SQNR	$\text{SQNR} = 3L^2 \dfrac{\overline{m^2}}{m_p^2}$	21-22 4(a)

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Subtopic	Formula	Equation	Used in
	Peak signal to peak noise	$\frac{S_{peak}}{N_{peak}} = L^2$	21-22 4(e)
	Minimum bandwidth	$B_T = \frac{R_b}{2} = nB$	21-22 4(e)
	Companded SQNR	$\text{SQNR} \approx \frac{3L^2}{[\ln(1+\mu)]^2}$	21-22 4(c)
Delta modulation (26)	DM update	$\hat{m}_q[k] = \hat{m}_q[k-1] + d_q[k],\ d_q = \pm\sigma$	19-20 4(e)
	DM receiver	$\hat{m}_q[k] = \sum_{i=0}^{k} d_q[i]$	19-20 4(e)
	DM largest slope	σf_s	20-21 3(a)
	No slope overload	$\left\lvert \frac{dm}{dt} \right\rvert_{max} \leq \sigma f_s$	20-21 3(a)
Sampling (12)	**Nyquist rate**	$f_s \geq 2B$	21-22 4(b), 20-21 4(c)
	Sampled spectrum	$\bar{G}(f) = \frac{1}{T_s}\sum_n G(f - nf_s)$	20-21 4(c)
	Interpolation formula	$g(t) = \sum_k g(kT_s)\,\text{sinc}(2\pi Bt - k\pi)$	21-22 4(b), 20-21 4(c)
	Sampled spectrum	$\bar{G}(f) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} G(f - nf_s)$	21-22 4(b)
	Reconstruction LPF	$H(f) = T_s\,\text{rect}\left(\frac{f}{2f_0}\right),\ B \leq f_0 \leq f_s - B$	21-22 4(b)

Symbols:

L = number of levels ↩

μ = compression constant (255), not the AM index ↩

x = fraction above Nyquist ↩

e = framing fraction ↩

n = bits per sample; bits per code word ↩

R_b = bit rate (bit/s) ↩

r = symbol rate (symbols/s) ↩

$\overline{m^2}$ = message power ↩

σ = step size ↩

B = highest message frequency (Hz) ↩

f_s = sampling rate (samples/s) ↩

$G(f)$ = message spectrum

$T_s = 1/f_s$ = sampling interval (s)

f_0 = filter cutoff (Hz) ↩

Questions:

No.	Paper	Marks	Type	Question
Quantization				
1	19-20 4(c)	4	theory	Define and explain uniform and non-uniform quantization.
PCM				
2	21-22 4(a)	5	theory	What is PCM? Explain PCM system.
3	21-22 4(c)	5	theory	Write short note on: Companding, Information rate, and Entropy of a source.
4	21-22 4(e)	5	numerical	An audio signal is sampled at 8 kHz rate using a 32-level quantizer. Find- (i) Peak signal power to peak quantization noise power. (ii) System bandwidth.

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No.	Paper	Marks	Type	Question
5	21-22 TT3 9(a)	7	numerical	It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patients are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitudes is 0.25% of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data.
6	20-21 3(c)	5	diagram	Explain the basic block diagram of PCM system.
7	19-20 4(d)	5	numerical	Five telemetry signals, each of bandwidth 2400 Hz, are to be transmitted simultaneously by binary PCM. The signals must be sampled at least 20% above the Nyquist rate. Framing and synchronizing requires an additional 0.5% extra bits. A PCM encoder is used to convert these signals before they are time-multiplexed into a single data stream. Determine the minimum possible data rate (bit per second) that must be transmitted, and the minimum bandwidth required to transmit the multiplexed signal.
Delta modulation				
8	21-22 5(a)	6	theory	Explain the principle of sigma-delta modulator. Also explain how it is advantageous over DM.
9	20-21 3(a)	5	theory	Define slope overload distortion and granular noise in delta modulation and how it is removed in ADM?
10	20-21 3(b)	5	diagram	Draw the transmitter and receiver circuits block diagram of delta modulation.
11	19-20 4(e)	5	derivation	Show that the receiver of DM is essentially an accumulator.
12	19-20 5(a)	5	theory	Explain principle of sigma-delta modulator. Also explain how it is advantageous over DM.
Sampling				
13	21-22 4(b)	5	theory	”A proper valued LPF filter can be used to reconstruct the sampled signal”-justify the statement with the proper mathematical explanations.
14	20-21 4(c)	7	derivation	State and prove Nyquist sampling theorem. Also find the interpolation formula for the reconstruction of analog signals from its uniform samples.

4.7. Digital modulation:

75 marks, 12 questions, in 4 of 4 papers. Book: Lathi Ch. 7, 10; Forouzan Ch. 5.

Most often theory (8 of 12). Subtopics: Line coding; DPSK; ASK, FSK and PSK; QPSK; FSK; QAM.

Formulas:

Subtopic	Formula	Equation	Used in
Line coding (7)	AMI	$0 \rightarrow 0$ V; 1s alternate $\pm V$	19-20 5(b)
	Dicode	$0 \rightarrow 1$: $-V$; $1 \rightarrow 0$: $+V$; no change: 0	19-20 5(b)
DPSK (22)	**DPSK encoder**	$q_k = I_k \oplus q_{k-1}$	22-23 8(b), 21-22 5(c), 20-21 6(d), 19-20 5(c)
	DPSK receiver	LPF output $= \frac{1}{2} a_k a_{k-1} A^2$	22-23 8(b), 19-20 5(c)
	DPSK receiver	$y_k \propto a_k a_{k-1}$	21-22 5(c), 20-21 6(d)
ASK, FSK and PSK (16)	**BFSK correlator**	$\int_0^{T_b} A \cos^2 \omega_1 t \, dt \approx \frac{AT_b}{2}$	19-20 5(d)

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Subtopic	Formula	Equation	Used in
	Coherent BFSK error	$P_b = Q\left(\sqrt{E_b/N_0}\right)$	19-20 5(d)
	ASK and PSK bandwidth	$B = (1+d)S$	21-22 4(d)
	Coherent BASK output	$dA\cos^2\omega_c t \to \frac{dA}{2}$	21-22 5(d)
QPSK (8)	**QPSK**	$\varphi = a_I A\cos\omega_c t + a_Q A\sin\omega_c t$	19-20 6(a)
	BPSK/QPSK error	$P_b = Q\left(\sqrt{2E_b/N_0}\right)$	19-20 6(a)
FSK (8)	FSK frequencies	$f_{1,0} = f_c \pm \Delta f$	20-21 6(a)
	FSK bandwidth	$B_{FSK} \approx 2\Delta f + 2R_b$	20-21 6(a)
	FSK bandwidth (Forouzan)	$B = (1+d)S + 2\Delta f$	21-22 4(d), 20-21 6(a)
QAM (14)	**QAM signal**	$\varphi_{QAM} = m_1\cos\omega_c t + m_2\sin\omega_c t$	20-21 6(b)
	QAM with phase error	$m_1\cos\theta - m_2\sin\theta$	20-21 6(b)
	QAM signal	$\varphi_{QAM}(t) = m_1(t)\cos\omega_c t + m_2(t)\sin\omega_c t$	21-22 5(b)
	QAM with phase error	$m_1(t)\cos\theta - m_2(t)\sin\theta$	21-22 5(b)

Symbols:

I_k = data bit ↩

q_k = encoded bit ↩

a_k = polar symbol (±1) ↩

y_k = LPF output ↩

S = signal (baud) rate; baud rate ↩

d = filter factor, $0 \le d \le 1$; data (1 or 0) ↩

R_b = bit rate ↩

Δf = half the spacing of f_1 and f_2 (Hz) ↩

m_1, m_2 = the two messages ↩

Questions:

No.	Paper	Marks	Type	Question
Line coding				
1	19-20 5(b)	7	diagram	Briefly explain the desired characteristics of a line coding scheme. Draw the line coded signal for the information bits '1010011100' for the following line coding scheme- i) Manchester, ii) AMI, iii) Di-code.
DPSK				
2	22-23 TT3 8(b)	4	diagram	Draw the block diagram of a Differential PSK (DPSK) modulator and demodulator and show how the I_k sequence [1010011100] changes in different parts of the block.
3	21-22 5(c)	6	numerical	Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010010110'.
4	20-21 6(d)	4	numerical	Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010011100'
5	19-20 5(c)	8	theory	Explain DPSK principle with necessary block diagram.
ASK, FSK and PSK				
6	21-22 4(d)	5	theory	Name three mechanisms for the modulation of digital data into an analog signal. Show the differences among them.
7	21-22 5(d)	6	theory	Briefly describe coherent and non-coherent detection of BASK and BFSK.
8	19-20 5(d)	5	theory	Describe coherent detection of BFSK.

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No.	Paper	Marks	Type	Question
QPSK				
9	19-20 6(a)	8	theory	With proper block diagram, describe modulation of QPSK. Also explain how it offers advantages over BPSK?
FSK				
10	20-21 6(a)	8	theory	Briefly describe FSK generation with the necessary block diagram. Include its spectrum and discuss the bandwidth requirements, along with the advantages and disadvantages of FSK.
QAM				
11	21-22 5(b)	7	theory	With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks.
12	20-21 6(b)	7	theory	With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks.

4.8. Multiplexing and multiple access:

82 marks, 15 questions, in 4 of 4 papers. Book: Lathi Ch. 11; Forouzan Ch. 6, 12.

Most often theory (9 of 15). Subtopics: TDM; FDM; Multiple access; Spread spectrum.

Formulas:

Subtopic	Formula	Equation	Used in
TDM (50)	**Frame rate**	$\frac{\text{channel rate}}{\text{bits per slot}}$	19-20 6(b)
	Frame and bit duration	$T_f = 1/\text{frame rate}$, $T_b = 1/R$	19-20 6(b)
	Link rate	$R = \text{frame rate} \times \text{bits per frame}$	19-20 6(b)
	Pulse stuffing	pad slow inputs to the fastest rate	19-20 6(d)
	Multiple slots	slots per frame $= R_i/R_{base}$	19-20 6(d)
	Quantization step	$\Delta = \frac{2m_p}{L}$	20-21 5(c)
	Largest error	$e_{max} = \frac{\Delta}{2} = \frac{m_p}{L}$	20-21 5(c)
	TDM bit rate	$R_b = N f_s n$	21-22 6(b), 20-21 5(c)
	Minimum bandwidth	$B_T = \frac{R_b}{2}$	20-21 5(c)
	Levels from an error limit	$\frac{m_p}{L} \le \epsilon\, m_p \;\Rightarrow\; L \ge \frac{1}{\epsilon}$	21-22 6(b)
	Synchronous TDM link	$R_{link} = \sum R_i$	21-22 6(a)
	Character TDM	frame $= 4 \times 8 + 1 = 33$ bits; link $= 250 \times 33 = 8250$ bit/s	21-22 9(c)
	Pulse stuffing	slow channel padded to the fast rate	22-23 8(c)
	TDM frame	frame rate = channel rate / bits per slot; R = frame rate × bits per frame	22-23 8(c)
FDM (10)	**FDM bandwidth**	$B_{total} = NB_{ch} + (N-1)B_g$	19-20 6(c)
	FDM link bandwidth	$B_{link} = NB_{ch} + (N-1)B_g$	20-21 2(b)
Spread spectrum (6)	FHSS channels	$M = 2^k$, $B_{SS} \approx MB$	20-21 6(c)

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Symbols:

m_p = peak signal ↩

N = number of channels ↩

ϵ = allowed error as a fraction of m_p ↩

R_i = input rates ↩

B_g = guard band ↩

k = PN pattern bits

Questions:

No.	Paper	Marks	Type	Question
TDM				
1	22-23 TT3 8(c)	6	numerical	Two channels, one with a bit rate of 190 kbps and another with a bit rate of 180 kbps, are to be multiplexed using pulse stuffing TDM with no synchronization bits. Answer the following questions: (a) What is the size of a frame in bits? (b) What is the frame rate? (c) What is the duration of a frame? (d) What is the data rate?
2	21-22 6(a)	6	theory	Write short notes on: Multilevel multiplexing and Multiple slot allocation.
3	21-22 6(b)	6	numerical	It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patients are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitudes is 0.25 % of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data.
4	21-22 TT3 9(b)	6	theory	Write short notes in multilevel multiplexing and multiple slot allocation.
5	21-22 TT3 9(c)	7	numerical	We have four sources, each creating 250 characters (8 bits per character) per second. If the interleaved unit is a character and 1 synchronizing bit is added to each frame, find (a) the data rate of each source, (b) the duration of each character in each source, (c) the frame rate, (d) the duration of each frame, (e) the number of bits in each frame, and (f) the data rate of the link.
6	20-21 5(c)	7	numerical	It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patents *[sic: should be "patients"]* are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitude is 0.25% of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data.
7	19-20 6(b)	6	numerical	A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output for the following inputs- '110010', '001010', '101101', '000111'. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?
8	19-20 6(d)	6	theory	Write short notes on following terms- (i) Pulse stuffing (ii) Multi-slot multiplexing
FDM				
9	20-21 2(b)	5	numerical	Five channels, each with 100 kHz bandwidth, are to be frequency division multiplexing together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channel to prevent interference?
10	19-20 6(c)	5	theory	Briefly explain frequency division multiplexing techniques.
Multiple access				
11	21-22 6(c)	3	theory	Compare between FDMA and CDMA.
12	21-22 6(e)	3	theory	Differentiate between multiplexing and multiple access technologies.
13	20-21 2(a)	5	theory	Differentiate between multiplexing and multiple access technologies.
14	20-21 5(a)	5	theory	Write down the comparison between TDMA and CDMA.

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No.	Paper	Marks	Type	Question
Spread spectrum				
15	20-21 6(c)	6	theory	Explain the principle of frequency hopping spread spectrum technique.

4.9. Information theory:

29 marks, 5 questions, in 3 of 4 papers. Book: Lathi Ch. 13.

Most often numerical (3 of 5). Subtopics: Entropy.

Formulas:

Subtopic	Formula	Equation	Used in
Entropy (29)	Information in a message	$I_i = \log_2 \frac{1}{P_i}$ bits	19-20 2(a)
	Entropy	$H(m) = \sum_{i=1}^{n} P_i \log_2 \frac{1}{P_i}$ bits/message	19-20 2(a)
	Count in a long sequence	$N_i = NP_i$	19-20 2(a)
	Information rate	$R = rH$ bits/s	19-20 2(a)
	Information in a symbol	$I = \log_2 \frac{1}{P}$ bits	20-21 1(a)
	Entropy	$H = \sum_i P_i \log_2 \frac{1}{P_i}$ bits/symbol	20-21 1(a)
	Binary entropy	$H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$	21-22 1(a), 20-21 1(a)
	Information in a symbol	$I(m_i) = \log_2 \frac{1}{P_i}$ bits	21-22 1(a)
	Entropy	$H(m) = \sum_{i=1}^{n} P_i \log_2 \frac{1}{P_i}$ bits/symbol	21-22 1(a)
	Bounds of entropy	$0 \leq H(m) \leq \log_2 n$	21-22 1(a)
	Rate of a four-message source	$H = 1.846$ bits, $R = rH = 1.846$ Mbit/s	21-22 7(a)

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No.	Paper	Marks	Type	Question

Symbols:

P_i = probability of message m_i; probability of symbol m_i ↩

N = number of messages sent

r = messages per second

P = probability of the symbol ↩

H = average information per symbol ↩

p = probability of one symbol; probability of one of two symbols ↩

n = number of symbols ↩

Questions:

No.	Paper	Marks	Type	Question
Entropy				
1	21-22 1(a)	6	numerical	Find the entropy $H(m)$ of a memoryless source.
2	21-22 TT1 7(a)	5	numerical	A message source generates one of the four messages randomly every microsecond. The probabilities of these messages are 0.4, 0.3, 0.2, and 0.1. Each emitted message is independent of the other messages in the sequence: (i) Find the source entropy. (ii) Find the rate of information generated by this source (in bits per second).
3	20-21 1(a)	5	numerical	Find the entropy of single bit binary memoryless source. Also draw the curve of entropy Vs. probability for that source.
4	20-21 TT1 7(a)	5	diagram	Find the entropy of single bit binary memoryless source. Also draw the curve of entropy vs probability for that source.
5	19-20 2(a)	8	derivation	Show that the average information per message is equal to the entropy of the information source.

4.10. System design:

7 marks, 1 question, in 1 of 4 papers. Book: Lathi Ch. 1.

Most often theory (1 of 1). Subtopics: Link design.

Questions:

No.	Paper	Marks	Type	Question
Link design				
1	21-22 6(d)	7	theory	Discuss the design procedures of a communication system.

5. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Signals and spectrum				
1	21-22 1(d)	6	Distortionless transmission	Write the output as a delayed scaled input; Fourier transform it; read magnitude and phase conditions.
2	20-21 TT1 7(b)	5	Distortionless transmission	Write the output as a delayed scaled input; Fourier transform it; read magnitude and phase conditions.
Noise and channel capacity				
3	21-22 TT1 7(b)	5	Cascaded noise	Model each stage as a noiseless gain with a noise resistor; add output noise powers; divide by the total squared gain.
4	20-21 TT1 7(d)	5	Cascaded noise	Model each stage as a noiseless gain with a noise resistor; add output noise powers; divide by the total squared gain.
5	19-20 1(d)	7	Cascaded noise	Model each stage as a noiseless gain with a noise resistor; add output noise powers; divide by the total squared gain.

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No.	Paper	Marks	Derivation	Outline
Amplitude modulation				
6	22-23 TT2 7(e)	0	DSB-SC	Multiply DSB-SC and SSB by a carrier with frequency and phase error; low-pass filter; read attenuation, beating, phase and frequency shift.
7	21-22 2(d)	6	VSB	Write the VSB spectrum; shift by the carrier; keep the baseband terms; require a constant bracket.
8	21-22 TT1 7(d)	5	Envelope detector	Compare the capacitor discharge slope with the envelope slope; find the worst instant; get the RC bound.
9	21-22 TT2 8(b)	6	SSB	Add a large carrier to SSB; write the envelope; expand the square root for a large carrier.
10	21-22 TT2 8(c)	7	VSB	Write the four DSB lines; apply the filter gains 1, a, 1-a, 0; demodulate; the partial sidebands add to the full tone.
11	20-21 4(b)	7	VSB	Shape DSB-SC with a filter; multiply by the carrier; filter; require the sum of the shifted responses to be constant.
12	20-21 TT2 8(b)	5	DSB-SC	Multiply by a carrier with frequency and phase error; low-pass filter; read the cos factor.
13	19-20 2(c)	6	DSB-SC	Multiply by a carrier with frequency and phase error; low-pass filter; read the cos factor.
14	19-20 3(a)	8	SSB	Two branches m cos and m_h sin; tone case with product identities; subtract or add to cancel a sideband.
Angle modulation				
15	22-23 TT2 7(i)	0	NBFM	Expand cos of the FM angle; use small-angle approximations; get carrier minus quadrature sideband term.
Sampling and pulse modulation				
16	20-21 4(c)	7	Sampling	Multiply by an impulse train; write its Fourier series; read off the repeated spectrum; filter with an ideal LPF.
17	19-20 4(e)	5	Delta modulation	Write the DM update; unroll it; the estimate is the running sum of the steps.
Information theory				
18	19-20 2(a)	8	Entropy	Count each message NP_i times in N messages; add their information; divide by N.

6. Repeat tracker:

No repeats among these papers.

7. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical	Diagram	Design
Signals and spectrum	3	2	·	·	·
Noise and channel capacity	1	3	5	·	·
Amplitude modulation	11	9	7	2	·
Angle modulation	7	1	5	1	2
Receivers	1	·	·	·	·
Sampling and pulse modulation	7	2	3	2	·
Digital modulation	8	·	2	2	·
Multiplexing and multiple access	9	·	6	·	·
Information theory	·	1	3	1	·
System design	1	·	·	·	·

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8. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Amplitude modulation	125	22 %
2	Angle modulation	85	37 %
3	Multiplexing and multiple access	82	52 %
4	Digital modulation	75	65 %
5	Sampling and pulse modulation	74	78 %
6	Noise and channel capacity	49	87 %
7	Signals and spectrum	30	92 %
8	Information theory	29	98 %
9	Receivers	7	99 %
10	System design	7	100 %

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